

SQL Cheat Sheet

1. Basic SELECT Queries

- SELECT * FROM employees;
- SELECT first_name, last_name, salary FROM employees;
- SELECT DISTINCT department_id FROM employees;

2. Filtering Data (WHERE Clause)

- SELECT * FROM employees WHERE department_id = 50;
- SELECT * FROM employees WHERE salary > 5000;
- SELECT * FROM employees WHERE hire_date BETWEEN '01-Jan-1999' AND '31-Dec-1999';
- SELECT * FROM employees WHERE job_id IN ('IT_PROG', 'SA_MAN');
- SELECT * FROM employees WHERE department_id NOT IN (50, 80);
- SELECT * FROM employees WHERE last_name LIKE 'M%'; -- Starts with M
- SELECT * FROM employees WHERE last_name LIKE '%o%'; -- Contains 'o'
- SELECT * FROM employees WHERE bonus IS NULL;
- SELECT * FROM employees WHERE bonus IS NOT NULL;

3. Sorting Results (ORDER BY)

- SELECT first_name, last_name, salary FROM employees ORDER BY salary;
- SELECT first_name, last_name, salary FROM employees ORDER BY salary DESC;

4. Subqueries (WHERE with Nested Queries)

- SELECT first_name, last_name, salary FROM employees WHERE salary > (SELECT salary FROM employees WHERE last_name = 'Davies');
- SELECT * FROM employees WHERE department_id = (SELECT department_id FROM employees WHERE last_name = 'Taylor');
- SELECT first_name, last_name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees WHERE department_id = 50);

5. String Functions

- SELECT LOWER(email) FROM employees WHERE department_id IN (50, 80);
- SELECT first_name || ' ' || last_name AS 'Full Name' FROM employees;
- SELECT UPPER(first_name) FROM employees WHERE manager_id = 124;
- SELECT SUBSTR(first_name, 1, 1) || last_name FROM employees;

6. Aggregate Functions (SUM, AVG, MIN, MAX, COUNT)

- SELECT AVG(salary) FROM employees;
- SELECT MAX(salary) FROM employees WHERE job_id IN ('IT_PROG', 'AD_VP');
- SELECT MIN(salary) FROM employees WHERE department_id IN (50, 80, 90);
- SELECT COUNT(*) FROM employees;

7. Mathematical Operations on Salary

- SELECT last_name, salary, salary + salary * 0.05 AS 'New Salary' FROM employees;
- SELECT first_name, last_name, salary, salary - salary * 0.03 AS 'Updated Salary' FROM employees;
- SELECT last_name, salary + 500, department_id FROM employees WHERE hire_date > '01-Jan-1998';

8. Substitution Variables (User Input in SQL Clients like SQL*Plus)

- SELECT * FROM employees WHERE job_id = :enter_job_id;
- SELECT first_name, last_name, department_id, salary FROM employees WHERE department_id = :enter_dept_id;

9. Grouping Data (GROUP BY and HAVING)

- SELECT department_id, AVG(salary) FROM employees GROUP BY department_id;
- SELECT department_id, AVG(salary) FROM employees GROUP BY department_id HAVING AVG(salary) > 5000;

10. Handling Dates

- SELECT * FROM employees WHERE hire_date > '01-Jan-2004';
- SELECT * FROM employees WHERE hire_date BETWEEN '01-Jan-2004' AND '31-Dec-2005';
- SELECT first_name, last_name, salary FROM employees ORDER BY hire_date DESC;

11. Advanced Subqueries

- SELECT first_name, last_name, salary FROM employees WHERE salary > (SELECT salary FROM employees WHERE last_name LIKE 'Davies');
- SELECT * FROM employees WHERE department_id = (SELECT department_id FROM employees WHERE last_name LIKE 'Davies');
- SELECT first_name, last_name, salary FROM employees WHERE salary < (SELECT salary FROM employees WHERE last_name LIKE 'Davies');
- SELECT first_name, last_name, salary FROM employees WHERE hire_date > (SELECT hire_date FROM employees WHERE last_name LIKE 'Davies');
- SELECT first_name, last_name, salary FROM employees WHERE manager_id = (SELECT manager_id FROM employees WHERE last_name LIKE 'Grant');
- SELECT first_name, last_name, salary FROM employees WHERE hire_date > (SELECT hire_date FROM employees WHERE last_name = 'Davies') AND salary < (SELECT AVG(salary) FROM employees WHERE hire_date > (SELECT

hire_date FROM employees WHERE last_name = 'Davies'));

- SELECT first_name, last_name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees WHERE department_id=50);

- SELECT first_name, last_name, salary FROM employees WHERE hire_date > (SELECT hire_date FROM employees WHERE last_name LIKE 'Matos') AND salary > (SELECT MIN(salary) FROM employees WHERE last_name LIKE 'Matos');

- SELECT first_name, last_name, salary, job_id, department_id FROM employees WHERE job_id = (SELECT job_id FROM employees WHERE last_name LIKE 'Matos');

- SELECT first_name, last_name, salary FROM employees WHERE manager_id = (SELECT manager_id FROM employees WHERE last_name LIKE 'Davies');

- SELECT first_name, last_name, salary FROM employees WHERE department_id = (SELECT department_id FROM employees WHERE last_name LIKE 'Taylor');

- SELECT first_name, last_name, salary, salary + 500 FROM employees WHERE hire_date > (SELECT hire_date FROM employees WHERE last_name = 'Abel');

- SELECT first_name, last_name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees WHERE manager_id=124);

- SELECT first_name, last_name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees WHERE manager_id IN (124, 149));

- SELECT first_name || ' ' || last_name || ' ' || salary AS 'Information' FROM employees WHERE hire_date < (SELECT hire_date FROM employees WHERE last_name LIKE 'Zlotkey');

- SELECT UPPER(first_name) || ' ' || UPPER(last_name) AS 'Name' FROM employees WHERE job_id = (SELECT job_id FROM employees WHERE last_name LIKE 'Matos');

- SELECT first_name, last_name, salary FROM employees WHERE department_id = (SELECT department_id FROM employees WHERE last_name = 'Abel') ORDER BY salary;

- SELECT first_name, last_name, salary, hire_date FROM employees WHERE hire_date > (SELECT hire_date FROM employees WHERE last_name = 'Abel') ORDER BY hire_date;

- SELECT first_name, last_name, salary, salary-200 FROM employees WHERE hire_date > (SELECT hire_date FROM employees WHERE last_name = 'Abel');

- SELECT first_name, last_name, salary FROM employees WHERE manager_id = (SELECT manager_id FROM employees WHERE last_name = 'Vargas');

- SELECT first_name, last_name, department_id FROM employees WHERE salary > (SELECT salary FROM employees WHERE first_name = 'Ellen' AND last_name = 'Abel') ORDER BY hire_date;

- SELECT first_name, last_name, salary FROM employees WHERE department_id = (SELECT department_id FROM employees WHERE first_name = 'Ellen') ORDER BY salary;

- SELECT first_name, last_name, salary, hire_date FROM employees WHERE hire_date > (SELECT hire_date FROM employees WHERE first_name = 'Ellen') ORDER BY salary;

- SELECT first_name, last_name, salary, hire_date FROM employees WHERE salary > (SELECT AVG(salary) FROM employees WHERE job_id IN ('ST_CLERK', 'AD_VP')) ORDER BY salary;

- SELECT first_name || ' ' || last_name AS 'Name' FROM employees WHERE job_id = (SELECT job_id FROM employees WHERE last_name LIKE 'Mourgos');
- SELECT first_name, last_name, salary FROM employees WHERE salary > (SELECT MIN(salary) FROM employees WHERE department_id=80);
- SELECT first_name, last_name, salary FROM employees WHERE department_id = (SELECT department_id FROM employees WHERE last_name LIKE 'Vargas') ORDER BY salary DESC;
- SELECT first_name, last_name, salary+salary*0.10, department_id FROM employees WHERE department_id = (SELECT department_id FROM employees WHERE last_name LIKE 'Vargas');
- SELECT first_name, last_name, salary, salary-salary*0.03 FROM employees WHERE hire_date > (SELECT hire_date FROM employees WHERE last_name LIKE 'Grant');
- SELECT last_name, salary+500, manager_id FROM employees WHERE manager_id = (SELECT manager_id FROM employees WHERE last_name LIKE 'Abel');
- SELECT first_name, last_name, salary, salary-300 FROM employees WHERE job_id = (SELECT job_id FROM employees WHERE last_name LIKE 'Davies');
- SELECT UPPER(first_name) FROM employees WHERE manager_id = (SELECT manager_id FROM employees WHERE last_name LIKE 'Mourgos');
- SELECT ROUND(AVG(salary),2) FROM employees WHERE department_id = (SELECT department_id FROM employees WHERE last_name LIKE 'Vargas');